

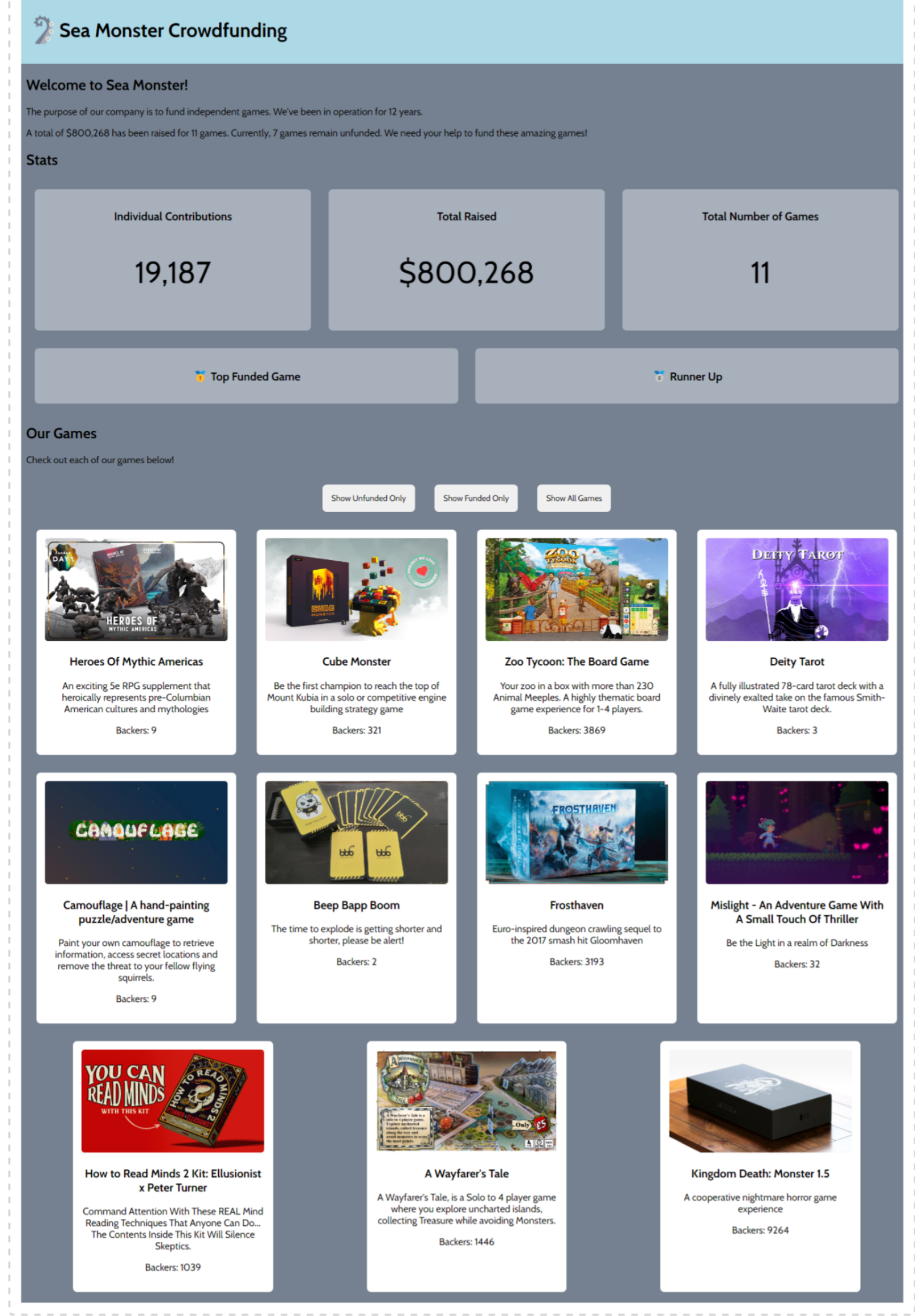
WEB102 Prework

Challenge 7 of 7: Select & display the top 2 games

✔ Checkpoint

Welcome to your final challenge! You're nearly ready to tackle React in Unit 1. We are legitimately so excited for you. There's just one more tool we need to get familiar with. Let's do it!

At this point, your website should look something like the screenshot below. Compare your work with the exemplar and see if there are any changes you might need to make!



🖋 Background

Destructuring

Imagine you're going to create a detailed info page for just one of the games in our list from Sea Monster. You have some data about the game from `GAMES_JSON`, but you don't need any other games right now. JavaScript gives us tools for "unpacking" objects and lists, allowing us to grab and rename elements for ease of use. Here's an example with our data from `GAMES_JSON`:

```
const { name, desc, pl, goal, bkrs, img } = GAMES_JSON[0];
```

Let's break that down. Instead of having to constantly refer to `GAMES_JSON[0].name`, we can now just say `name`. We unpacked all the values associated with "Heroes of Mythic Americas" to make creating the info page way easier.

This operation is called "destructuring" and it can be used in a variety of ways. Here are some other quick examples:

```
let a, b = [10, 20];

let func1, func2 = [(arg) => arg * 2, (arg) => arg * 3]

function getEmployees() {
  return ["Sandra", "Ola", "Chi"];
}
let {a, b, c} = getEmployees();

var a, b;
[a, , b] = ["Roan", "Bay", "Dappled"];
```

Spread

Now imagine we're back at Spotify, this time working on showing each user their least played song. We have a sorted list of `songs` and our job is to grab the last element. Here's one way we could do that:

```
let [...others, leastPlayed] = songs;
```

What just happened there? The variable `others` now contains every song before the least played song, and `leastPlayed` contains the last element of `songs`. The `...` is called the "spread" operator, and it allows us to quickly copy all or part of an existing array or object into another array or object.

Here's one useful way to copy over elements from two arrays:

```
const numbersOne = [1, 2, 3];
const numbersTwo = [4, 5, 6];
const numbersCombined = [...numbersOne, ...numbersTwo];
// => [1, 2, 3, 4, 5, 6]
```

Another very common use for the spread operator that you will see in React is to add a new element to an existing list:

```
let songs = ["Song 1", "Song 2", "Song 3"];
let newSongs = [...songs, "Song 4"];
// => ["Song 1", "Song 2", "Song 3", "Song 4"]
```

Why not just use something like `.push()` to add the element? While that is a very common way to add an element to an array, it mutates the existing list instead of creating a new one. React is sometimes waiting for a new list to be created before making changes to the user interface. More on this later, of course!

🎯 Challenge Steps

➡️➡️➡️ Step 1: Calculating the Top Games

Use destructuring and the spread operator to grab the first and second games from the `sortedGames` list. These games will be the top two most funded games in the list.

🔑 **Secret Key component 1:** What is the **first word** of the most funded game?

🔑 **Secret Key component 2:** What is the **first word** of the second most funded game?

➡️➡️➡️ Step 2: Displaying the Top Games

Create a new element that contains the name of the top funded game and appends it to the `firstGameContainer`. Do the same for the second most funded game, appending it to the `secondGameContainer`.

🔑 **Secret Key component 3:** What is the value of `...rest` in the following code segment?

```
const numbers = [1, 2, 3, 4, 5, 6];
const [one, two, ...rest] = numbers;
```

1. `null` (keyword: CRABAPPLE)
2. `[1, 2, 3, 4, 5, 6]` (keyword: MAPLE)
3. `[]` (keyword: ASH)
4. `[3, 4, 5, 6]` (keyword: CEDAR)

➡️➡️➡️ Step 3: Customizations

Amazing work on the site! You've completed all the technical steps to get Sea Monster Crowdfunding's website up and running with all the required features.

The website is relatively sparse at the moment. Take a moment to consider what features you think would help the site stand out more? Take a moment to sketch out some improvements to the site, whether they are to its look and feel or to its functionality. Some examples may include:

- What if a user discovers the site and wants to search for a specific game they have heard of?
- What if a user wants to use the nav bar at the top of the page to get to the Our Game section quickly?
- How would you update the CSS of the site to make it more visually appealing?

Can you implement any of these features? You've got all the skills you need to make big upgrades to Sea Monster Crowdfunding's landing page. Give it a try! See if you can implement one bonus feature to the site.

🚀 Moving on to the Submission Instructions

Create the 🔑 **Secret Key** from its components with no spaces in between:

- Component 1: What is the **first word** of the most funded game?
- Component 2: What is the **first word** of the second most funded game?
- Component 3: What is the value of `...rest` in the following code segment?

Hint: your 🔑 **Secret Key** should be 11 characters long.

Reminders about 🔑 **Secret Keys**:

- Keys for the Offline PDF version of prework **are case sensitive!**
 - (For example, "SEA_MONSTER" is NOT the same as "sea_monster")
- Secret keys can be any combination of letters, numbers, and symbols.

Use your 🔑 **Secret Key** to unlock the next PDF file. If your 🔑 **Secret Key** is correct, you will move on to the submission instructions!